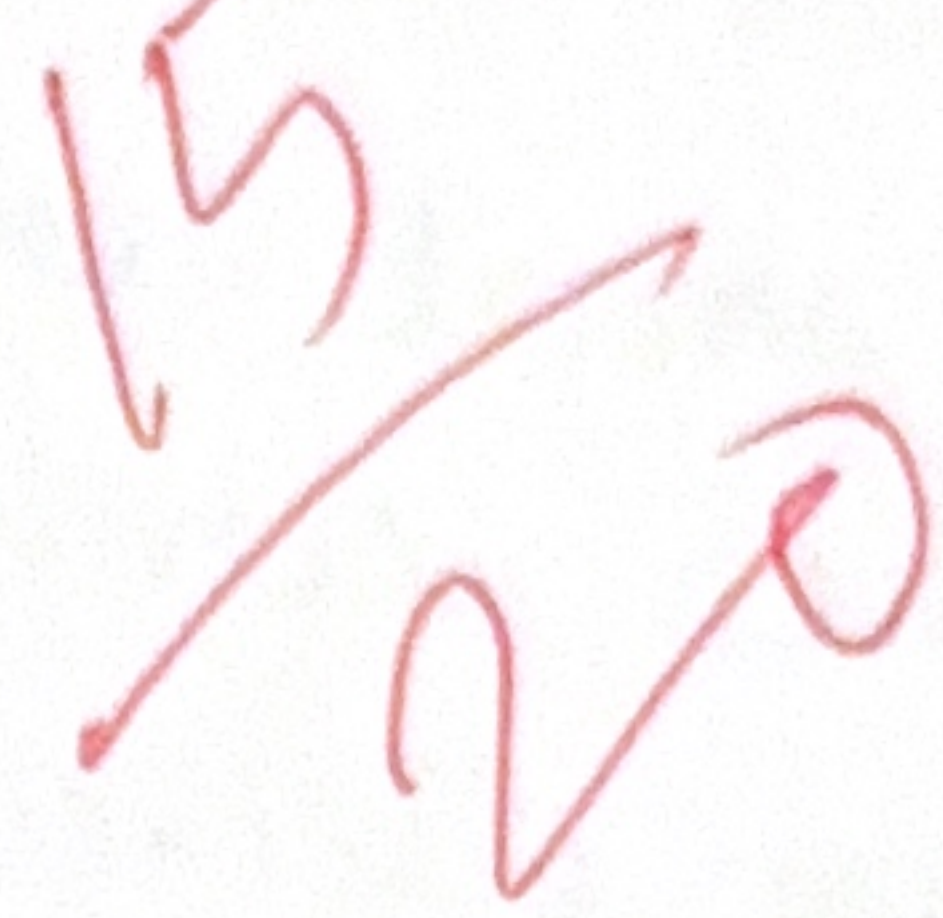


i) Intelligent Agents in a smart Home Automation System:

An intelligent system that perceives its environment through sensors and takes action through actuators to achieve a desired object.

In a smart home, sensors may detect

- Motion or human presence.
- Temperature
- light level
- Time of day.
- Door/window status
- User activities and preferences.



The agent then controls devices such as lights, fans, air conditioners.

1. Simple Reflex Agent:

A simple reflex agent chooses an action based only on the current percept using condition-action rule.

Structure:

If condition $\rightarrow$ perform action

2. Model-Based Agent:

A model based agent maintains an internal model / state of the environment. It uses current sensor information together with information about previous situations.

3. Goal-Based Agent:

A goal based agent selects actions based on a desired goal.

For a smart home, goals could include:

1. Maintain comfortable
2. Reduce electricity consumption
3. Keep the house secure.
4. Turn off unnecessary appliances

4. Utility-Based Agent:

A utility-based agent chooses the action that provides the highest utility, meaning the best overall outcome according to several criteria:

1. Comfort
2. Energy efficiency
3. Security
4. Costs

28] Robot Solving an Unknown Maze Using UCS and IDS

Consider a robot placed in an unknown maze. The robot does not know the maze layout or the location of the exit beforehand. It must explore different paths until it finds a path to goal.

This is a search problem where:

1. Initial state $\rightarrow$ Robot's starting position.
2. Actions $\rightarrow$ Move up, down, left, right.
3. States $\rightarrow$ Different positions in the maze.
4. Goal State $\rightarrow$ Exit of the maze.
5. Path cost $\rightarrow$ cost of movements.

Two uniformed search strategies can be used are

1. Uniform Cost Search.
2. Iterative Deepening Search.

Uniform Cost Search:

Uniform cost search expands the node with the lowest total path cost.

It uses a priority queue.

If the robot has two possible paths.

- Path A = cost 5
- Path B = cost 10.

As UCS explores Path A first

Iterative Deepening Search (IDS):

Iterative Deepening search repeatedly performs depth-limited search, increasing the depth limit each time.

For example:

Depth 0 → Depth 1 → Depth 2 → Depth 3

Example:

Suppose the exit is 3 moves away.

IDS

- Depth 0
- Depth 1
- Depth 2
- Depth 3 → Exit found.